Written exam QM1 channel B - 10/02/2025

The candidate must choose **three exercises** *out of the four presented in the following.* **Do not** *submit a text with more than three exercises, otherwise only the first three ones, in the order assigned by the candidate, will be taken into account. On each sheet, please indicate* **first and last name**, *and the university* **registration number**.

Exercise 1

A quantum two-level system is governed by the following hamiltonian operator:

$$\mathcal{H} = -\frac{1}{2}\hbar\omega\big(|0\rangle\langle 0| - |1\rangle\langle 1|\big)\,, \tag{1}$$

where $|0\rangle$ and $|1\rangle$ are its orthonormal eigenstates. As a first step, determine the corresponding eigenvalues. Then, consider the operator $a = |0\rangle\langle 1|$ and its hermitian conjugate $a^\dagger$.

a) Evaluate $\{a, a^\dagger\} \equiv aa^\dagger + a^\dagger a$, a^2, $a^{\dagger 2}$, $\big[H, a\big]$, $\big[H, a^\dagger\big]$.

b) Assuming that at $t = 0$ the system is described by the eigenstate of the hermitian operator $A = a + a^\dagger$ with eigenvalue 1, calculate the expectation valus of A, A^2 and the variance of A as a function of time t.

c) Let's consider another hermitian operator $B = -i\big(a - a^\dagger\big)$. Calculate the expectation value of B, B^2 and the variance of B as a function of time, using the same state as in b). Check the validity of the uncertainty relation between the operators A and B.

Exercise 2

A free particle has the initial wave function

$$\psi(x, 0) = A\, e^{-a\, x^2}\,, \tag{2}$$

where A and a are real and positive constants.

a) Normalize $\psi(x, 0)$.

b) Determine the wavefunction $\psi(x, t)$ with $t > 0$.

c) Find the probability density as a function of x and t. Express the answer in terms of the quantity

$$\omega = \sqrt{a/[1 + (2\hbar a t/m)^2]}\,. \tag{3}$$

Draw the probability density as a function of x at $t = 0$, and again for some $t > 0$. Qualitatively, what happens to the probability density as the time goes on?

Exercise 3

A simple harmonic oscillator in one dimension is subject to the perturbation:

$$H' = b\,x\,, \tag{4}$$

where b is a real constant.

a) Calculate the energy shift of the ground state to the lowest non-vanishing order.

b) Solve the problem exactly and compare the result with what can be obtained using perturbation theory (Hint: there's a much simpler solution than calculating explicitly the energy spectrum).

Exercise 4

The spin-dependent hamiltonian of an electron-positron (e^-e^+) system in the presence of a uniform magnetic field B in the z-direction can be written as

$$H = A\,\mathbf{S}^{(e^-)} \cdot \mathbf{S}^{(e^+)} + \left(\frac{eB}{mc}\right)\left(S_z^{(e^-)} - S_z^{(e^+)}\right). \tag{5}$$

where A is a constant. Let us also assume that the state of the system is given by the following product of spinor states:

$$\chi_+^{(e^-)}\,\chi_-^{(e^+)}\,, \tag{6}$$

where the plus and minus subscripts refer respectively to the up and down configurations along the z direction.

a) Is this an eigenstate of H in the limit $A \to 0$, $B \neq 0$? If it is, what is the corresponding energy eigenvalue? If it is not, what is the expectation value of H?

b) Answer the same questions, but in the limit $B \to 0$, $A \neq 0$

Useful formulas

Gaussian integral (with $a > 0$):

$$\int_{-\infty}^{+\infty} dx\, e^{-ax^2} = \sqrt{\frac{\pi}{a}} \tag{7}$$

Integrals of the form

$$\int_{-\infty}^{+\infty} dx\, e^{-(ax^2+bx)} \tag{8}$$

can be handled by completing the square: let $y \equiv \sqrt{a}\,[x + (b/2a)]$ and note that $(ax^2 + bx) = y^2 - b^2/4a$.

{ Exercise 1 }   WRITTEN EXAM, 10/02/2025

$$H = -\frac{\hbar\omega}{2}\left(|0\rangle\langle 0| - |1\rangle\langle 1|\right) \quad , \; |0\rangle, |1\rangle \text{ orthonormal}$$

$$H|0\rangle = -\frac{\hbar\omega}{2}|0\rangle , \quad H|1\rangle = \frac{\hbar\omega}{2}|1\rangle$$

$$a = |0\rangle\langle 1|$$

$$a^\dagger = |1\rangle\langle 0|$$

a) $$\{a, a^\dagger\} = aa^\dagger + a^\dagger a = |0\rangle\overbrace{\langle 1|1\rangle}^{1}\langle 0| + |1\rangle\overbrace{\langle 0|0\rangle}^{1}\langle 1|$$

$$= |0\rangle\langle 0| + |1\rangle\langle 1| = \mathbb{1}$$

$$a^2 = |0\rangle\underbrace{\langle 1|0\rangle}_{0}\langle 1| = 0$$

$$a^{\dagger 2} = |1\rangle\overbrace{\langle 0|1\rangle}^{0}\langle 0| = 0$$

$$[H, a] = -\frac{\hbar\omega}{2}\left(|0\rangle\underbrace{\langle 0|0\rangle}_{1}\langle 1| - \cancel{|1\rangle\langle 1|0\rangle}\langle 1|\right) +$$

$$+\frac{\hbar\omega}{2}\left(|0\rangle\cancel{\langle 1|0\rangle}\langle 0| - |0\rangle\overbrace{\langle 1|1\rangle}^{1}\langle 1|\right) =$$

$$= -\frac{\hbar\omega}{2}\left(|0\rangle\langle 1| + |0\rangle\langle 1|\right) = -\hbar\omega\, a$$

$$[H, a^\dagger] = -\frac{\hbar\omega}{2}\left(|0\rangle\cancel{\langle 0|1\rangle}\langle 0| - |1\rangle\langle 1|1\rangle\langle 0|\right) +$$

$$+\frac{\hbar\omega}{2}\left(|1\rangle\langle 0|0\rangle\langle 0| - |1\rangle\cancel{\langle 0|1\rangle}\langle 1|\right) =$$

$$= +\frac{\hbar\omega}{2}\left(|1\rangle\langle 0| + |1\rangle\langle 0|\right) = \hbar\omega\, a^\dagger$$

b) $A = a + a^\dagger = |0\rangle\langle 1| + |1\rangle\langle 0|$, $A = A^\dagger$

Find eigenstate of A with eigenvalue 1:

$$A|0\rangle = |1\rangle$$

$$A|1\rangle = |0\rangle$$

$$A\left(|0\rangle + |1\rangle\right) = \left(|0\rangle + |1\rangle\right)$$

$\Rightarrow |\psi\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + |1\rangle\right)$: normalized eigenstate of A with eigenvalue 1

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}}\left(e^{-i\frac{E_0}{\hbar}t}|0\rangle + e^{-i\frac{E_1}{\hbar}t}|1\rangle\right) =$$

$$= \frac{1}{\sqrt{2}}\left(e^{i\omega t/2}|0\rangle + e^{-i\omega t/2}|1\rangle\right)$$

$$\langle A\rangle = \langle\psi(t)|A|\psi(t)\rangle =$$

$$= \frac{1}{2}\left(\langle 0|e^{-i\omega t/2} + \langle 1|e^{i\omega t/2}\right)\left(|0\rangle\langle 1| + |1\rangle\langle 0|\right)\cdot$$

$$\cdot\left(|0\rangle e^{i\omega t/2} + |1\rangle e^{-i\omega t/2}\right) =$$

$$= \frac{1}{2}\left(e^{-i\omega t} + e^{i\omega t}\right) = \cos\omega t$$

$$\langle A^2 \rangle = \langle (a+a^\dagger)(a+a^\dagger) \rangle =$$

$$= \langle (\underbrace{a^2}_{0} + aa^\dagger + a^\dagger a + \underbrace{a^{\dagger 2}}_{0}) \rangle = \langle \overbrace{\{a, a^\dagger\}}^{1} \rangle =$$

$$= \langle \psi(t) | \psi(t) \rangle = 1$$ (normalization preserved by time evolution)

$$\Delta A^2 = \langle A^2 \rangle - \langle A \rangle^2 = 1 - \cos^2 wt = \text{sen}^2 wt$$

c) $B = -i(a - a^\dagger)$, $B = B^\dagger$

$$\langle B \rangle = \langle \psi(t) | B | \psi(t) \rangle =$$

$$= -\frac{i}{2} \left(\langle 0 | e^{-iwt/2} + \langle 1 | e^{iwt/2} \right) \left(|0\rangle\langle 1| - |1\rangle\langle 0| \right) \cdot$$

$$\cdot \left(|0\rangle e^{iwt/2} + |1\rangle e^{-iwt/2} \right) =$$

$$= -\frac{i}{2} \left(e^{-iwt} - e^{iwt} \right) = \frac{i^2}{2} 2 \, \text{sen}\, wt = -\text{sen}\, wt$$

$$\langle B^2 \rangle = \langle i^2 (a - a^\dagger)(a - a^\dagger) \rangle =$$

$$= -\langle (\underbrace{a^2}_{0} - aa^\dagger - a^\dagger a + \underbrace{a^{\dagger 2}}_{0}) \rangle = +\langle \{a, a^\dagger\} \rangle = 1.$$

$$\Delta B^2 = \langle B^2 \rangle - \langle B \rangle^2 = 1 - \text{sen}^2 wt = \cos^2 wt$$

$$\Delta A \cdot \Delta B \geq \frac{1}{2}\left|\langle \underbrace{[A,B]}_{=0} \rangle\right|$$

The relation is thus satisfied

$$\Delta A = \operatorname{sen} \omega t$$

$$\Delta B = \cos \omega t$$

$$[A,B] = \left[a+a^\dagger, -i(a-a^\dagger)\right] =$$

$$= -i\left\{ \cancel{[a,a]} + [a,-a^\dagger] + [a^\dagger,a] + \cancel{[a^\dagger,-a^\dagger]} \right\} =$$

$$= -i\left\{ -[a,a^\dagger] + [a^\dagger,a] \right\} = 2i\,[a,a^\dagger] =$$

$$= 2i\left(|0\rangle\underbrace{\langle 1|1\rangle}_{1}\langle 0| - |1\rangle\underbrace{\langle 0|0\rangle}_{1}\langle 1| \right) =$$

$$= 2i\left(-\frac{2}{\hbar\omega} H\right) = -\frac{4i}{\hbar\omega} H$$

$$\langle [A,B] \rangle = -\frac{4i}{\hbar\omega}\langle \psi(t)| H |\psi(t)\rangle =$$

$$= 2i\left(\langle 0|e^{-i\omega t/2} + \langle 1|e^{i\omega t/2}\right)\left(|0\rangle\langle 0| - |1\rangle\langle 1|\right) \cdot$$

$$\cdot \left(|0\rangle e^{i\omega t/2} + |1\rangle e^{-i\omega t/2}\right) =$$

$$= 2i(1-1) = 0$$

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{ Exercise 2 }

Free particle with wavefunction:

$$\psi(x, t=0) = A e^{-ax^2} \quad , \quad A, a \in \mathbb{R}^+$$

(Gaussian wavepacket)

a) $$1 = \int_{-\infty}^{+\infty} dx \, |\psi(x,0)| = A^2 \int_{-\infty}^{+\infty} dx \, e^{-2ax^2} =$$

$$= A^2 \sqrt{\frac{\pi}{2a}} \quad \Rightarrow \quad A = \left(\frac{2a}{\pi}\right)^{1/4}.$$

b) $\psi(x, t>0) = ?$

Write $\psi(x,0)$ using the free particle $\hat{H}$ basis:

$$\psi(x,0) = \int \frac{dk}{\sqrt{2\pi}} \phi(k) \, e^{ikx} \quad \rightarrow \text{ determine } \phi(k)$$

$$\phi(k) = \int_{-\infty}^{+\infty} \frac{dx}{\sqrt{2\pi}} \psi(x,0) \, e^{-ikx} =$$

$$= \frac{A}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} dx \, e^{-ax^2} e^{-ikx} =$$

complete the square
to use Gaussian
integrals

$$= \frac{A}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} dx \; e^{-(ax^2 + ikx)} =$$

$$\left[\begin{aligned} ax^2 + ikx &= a\left(x^2 + 2\frac{ik}{2a}x + \frac{i^2k^2}{4a^2}\right) - \frac{i^2k^2}{4a} = \\ &= a\left(x + \frac{ik}{2a}\right)^2 + \frac{k^2}{4a} = \\ &= ay^2 + k^2/4a \quad , \quad y = x + \frac{ik}{2a} \end{aligned}\right.$$

$$= \frac{A}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} dy \; e^{-ay^2} e^{-k^2/4a} =$$

$$= \frac{1}{\sqrt{2\pi}} \left(\frac{2a}{\pi}\right)^{1/4} e^{-\frac{k^2}{4a}} \sqrt{\frac{\pi}{a}} = e^{-\frac{k^2}{4a}} \frac{1}{(2a\pi)^{1/4}}$$

$$\Rightarrow \quad \phi(k) = \frac{e^{-k^2/4a}}{(2a\pi)^{1/4}}$$

$$\Rightarrow \quad \psi(x,0) = \int_{-\infty}^{+\infty} \frac{dk}{\sqrt{2\pi}} \phi(k) \, e^{ikx}$$

energy eigenvalue of free particle

$$E_k = \frac{\hbar^2 k^2}{2m}$$

$$\psi(x,t) = \int_{-\infty}^{+\infty} \frac{dk}{\sqrt{2\pi}} \phi(k) \, e^{-\frac{i}{\hbar}E_k t} \, e^{ikx} =$$

$$= \int_{-\infty}^{+\infty} \frac{dk}{\sqrt{2\pi}} \phi(k)\, e^{-\frac{i}{\hbar} E_k t}\, e^{ikx} =$$

$$= \int_{-\infty}^{+\infty} \frac{dk}{\sqrt{2\pi}} \frac{e^{-k^2/4a}}{(2a\pi)^{1/4}}\, e^{-i\frac{\hbar k^2}{2m}t}\, e^{ikx} = \circledast$$

→ let's complete the square again:

$$-\frac{k^2}{4a} - k^2 \frac{i\hbar t}{2m} + ikx =$$

$$= -\underbrace{\left(\frac{1}{4a} + \frac{i\hbar t}{2m}\right)}_{\beta} k^2 + ikx = -\beta\left(k^2 - \frac{ikx}{\beta}\right) =$$

$$= -\beta\left(k^2 - \frac{2ix}{2\beta}k + \frac{i^2x^2}{4\beta^2}\right) + \frac{i^2x^2}{4\beta} =$$

$$= -\beta\underbrace{\left(k - \frac{ix}{2\beta}\right)}_{k'}{}^2 - \frac{x^2}{4\beta} = -\beta k'^2 - \frac{x^2}{4\beta}$$

$$\circledast = \frac{1}{\sqrt{2\pi}} \frac{1}{(2a\pi)^{1/4}} e^{-x^2/4\beta} \int_{\mathbb{R}} dk'\, e^{-\beta k'^2} =$$

$$= \frac{1}{\sqrt{2\pi}} \frac{1}{(2a\pi)^{1/4}} e^{-x^2/4\beta} \sqrt{\frac{\pi}{\beta}} =$$

$$= \frac{1}{\sqrt{2\pi}} \frac{1}{(2a\pi)^{1/4}} e^{-x^2/4\beta} \sqrt{\frac{\pi}{\beta}} =$$

$$\left\{ \beta = \frac{1}{4a} + \frac{i\hbar t}{2m} = \frac{m + 2ai\hbar t}{4am} = \frac{1 + \frac{2ai\hbar t}{m}}{4a} \right\}$$

$$= \frac{1}{\sqrt{2}} \frac{1}{(2a\pi)^{1/4}} \frac{2\sqrt{a}}{\sqrt{1 + 2ia\hbar t/m}} e^{-x^2 4a / 4(1+2ia\hbar t/m)} =$$

$$= \frac{(2a)^{1/4}}{\pi^{1/4}} \frac{1}{\sqrt{1 + 2ia\hbar t/m}} e^{-ax^2/(1+2ia\hbar t/m)}$$

$$\|$$

$$\psi(x,t)$$

$$\gamma(t) \doteq 2a\hbar t/m$$

$\rightarrow$ so the wavefunction will flatten out over time

c) $$|\psi(x,t)|^2 = \left(\frac{2a}{\pi}\right)^{1/2} \frac{1}{\sqrt{(1+i\gamma)(1-i\gamma)}} \cdot$$

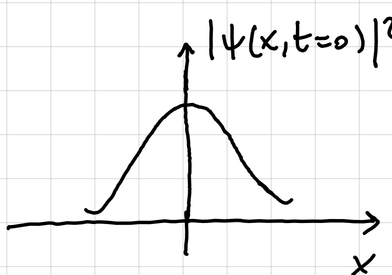
$$\cdot\, e^{-ax^2/(1+i\gamma)} \, e^{-ax^2/(1-i\gamma)} \, , \longrightarrow$$

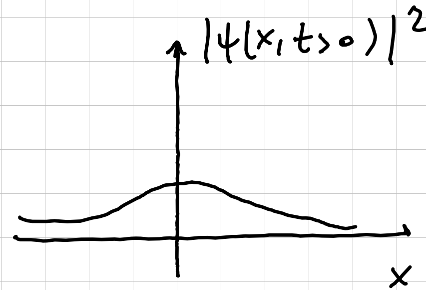
$$-\frac{ax^2}{1+i\gamma} - \frac{ax^2}{1-i\gamma} = \frac{-ax^2(1-i\gamma+1+i\gamma)}{1+\gamma^2} = \frac{-2ax^2}{1+\gamma^2}$$

$$|\psi(x,t)|^2 = \sqrt{\frac{2a}{\pi}}\, \frac{1}{\sqrt{1+\gamma^2}}\, e^{-2ax^2/(1+\gamma^2)}$$

$$w(t) = \left\{ \frac{a}{1+\gamma(t)^2} \right\}^{1/2}$$, decreasing function of time

$$|\psi(x,t)|^2 = \sqrt{\frac{2}{\pi}}\, w(t)\, e^{-2w(t)^2 x^2}$$





the wave packet broadens over time.

$\{$ Exercise 3 $\}$

Harmonic oscillator, 1D

perturbation $H' = b\hat{x}$, $b \in \mathbb{R}^+$

a) calculate the energy shift of the ground state to the lowest non-vanishing order

b) solve the problem exactly and compare with the results obtained in a)

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a)

$\{|n\rangle\}$, orthonormal, complete set of H eigenstates

$E_n = (n + \frac{1}{2})\hbar\omega$

$$E_0^{(1)} = \langle 0|H'|0\rangle = b\langle 0|\hat{x}|0\rangle$$

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}\,(\hat{a} + \hat{a}^{+})$$

$$a|n\rangle = \sqrt{n}\,|n-1\rangle$$

$$a^{+}|n\rangle = \sqrt{n+1}\,|n+1\rangle$$

$$E_0^{(1)} = b\sqrt{\frac{\hbar}{2m\omega}}\left\{\overbrace{\langle 0|\hat{a}|0\rangle}^{0} + \overbrace{\langle 0|a^{+}|0\rangle}^{\sim\langle 0|1\rangle = 0}\right\} = 0.$$

$$E_0^{(2)} = \sum_{k\neq 0} \frac{|\langle k|H'|0\rangle|^2}{E_0 - E_k} =$$

$$= \sum_{k\neq 0} \frac{b^2 |\langle k|\hat{x}|0\rangle|^2}{\cancel{\frac{1}{2}\hbar\omega} - \cancel{\frac{1}{2}\hbar\omega} - k\hbar\omega} = -\frac{b^2}{\hbar\omega} \sum_{k\neq 0} \frac{|\langle k|\hat{x}|0\rangle|^2}{k} = (*)$$

$$\rightarrow \langle k|\hat{x}|0\rangle = \sqrt{\frac{\hbar}{2m\omega}} \langle 0|\hat{a}^k (\hat{a}+\hat{a}^\dagger)|0\rangle =$$

$$= \sqrt{\frac{\hbar}{2m\omega}} \left\{ \langle 0|\hat{a}^k \underbrace{\hat{a}|0\rangle}_{0} + \langle 0|\hat{a}^k a^\dagger|0\rangle \right\} =$$

$$= \sqrt{\frac{\hbar}{2m\omega}} \langle 0|\hat{a}^k|1\rangle = \sqrt{\frac{\hbar}{2m\omega}}\, \delta_{k1}$$

$$(*) = -\frac{b^2}{\hbar\omega} \frac{\hbar}{2m\omega} \sum_{k\neq 0} \frac{\delta_{k1}}{k} = -\frac{b^2}{2m\omega^2}$$

$$\Rightarrow E_0 = \frac{\hbar\omega}{2} - \frac{b^2}{2m\omega^2}$$ $\rightarrow$ second order correction in b to E_0.

b) Find the exact solution:

$$\hat{H} = \hat{H}_0 + \hat{H}' = \frac{\hat{p}^2}{2m} + \frac{1}{2} m\omega^2 \hat{x}^2 + b\hat{x} =$$

$$= \frac{\hat{p}^2}{2m} + \frac{1}{2} m\omega^2 \hat{x}^2 + b\hat{x} =$$

completing the square one obtains:

$$= \frac{\hat{p}^2}{2m} + \frac{1}{2} m\omega^2 \left(\hat{x}^2 + \frac{2b}{m\omega^2}\hat{x}\right) =$$

$$= \frac{\hat{p}^2}{2m} + \frac{1}{2} m\omega^2 \left(\hat{x}^2 + 2\frac{b}{m\omega^2}\hat{x} + \frac{b^2}{m^2\omega^4}\right) - \frac{b^2}{2m\omega^2} =$$

$$= \frac{\hat{p}^2}{2m} + \frac{1}{2} m\omega^2 \hat{y}^2 - \frac{b^2}{2m\omega^2} \quad , \quad \hat{y} = \left(\hat{x} + \frac{b}{m\omega^2}\right)$$

which is the hamiltonian for a harmonic oscillator with shifted position $\hat{y}$ and an overall energy shift of $-b^2/2m\omega^2$, which coincides with the result found with second order perturbation theory.

{ Exercise 4 }

- $e^- e^+$ system (non-identical, spin $\frac{1}{2}$ fermions)
- B field in the $\hat{z}$ direction
- hamiltonian :

$$\hat{H} = A\, \hat{\vec{S}}^{(e^-)} \cdot \hat{\vec{S}}^{(e^+)} + \left(\frac{eB}{mc}\right)\left(\hat{S}_z^{(e^-)} - \hat{S}_z^{(e^+)}\right),$$

A, B, m, e, c : real constants

- State of the system : $\chi_+^{(e^-)} \chi_-^{(e^+)}$

(up $\equiv$ + and down $\equiv$ − states along $\hat{z}$)

one can also write : $|\psi\rangle = |\uparrow_z^{(e^-)} \downarrow_z^{(e^+)}\rangle \equiv$
$= |\uparrow\downarrow\rangle$

a) Is $|\psi\rangle$ an eigenstate of $H(A\to 0, B \neq 0)$?

$$\frac{eB}{mc}\left(\hat{S}_z^{(e^-)} - S_z^{(e^+)}\right)|\psi\rangle =$$

$- \frac{\hbar}{2}|\uparrow\downarrow\rangle$

$$= \frac{eB}{mc}\left(\hat{S}_z^{(e^-)}|\uparrow_z^{(e^-)}\downarrow_z^{(e^+)}\rangle - \hat{S}_z^{(e^+)}|\uparrow_z^{(e^-)}\downarrow_z^{(e^+)}\rangle\right) =$$

$+ \frac{\hbar}{2}|\uparrow\downarrow\rangle$

$$= \frac{eB}{mc}\left(\frac{\hbar}{2}\left|\uparrow_z^{(e^-)}\downarrow_z^{(e^+)}\right\rangle + \frac{\hbar}{2}\left|\uparrow_z^{(e^-)}\downarrow_z^{(e^+)}\right\rangle\right) =$$

$$= \frac{eB}{mc}\hbar\,|\psi\rangle$$

$\Rightarrow$ $|\psi\rangle$ is eigenstate of $H(A\to 0, B\neq 0)$ with eigenvalue $\frac{eB\hbar}{mc}$.

b) $\hat{H}(A\neq 0, B\to 0)|\psi\rangle = ?$

$$A\,\hat{\vec{S}}^{(e^-)}\cdot\hat{\vec{S}}^{(e^+)}\left|\uparrow_z^{(e^-)}\downarrow_z^{(e^+)}\right\rangle \neq \text{cost.}\left|\uparrow_z^{(e^-)}\downarrow_z^{(e^+)}\right\rangle$$

because the eigenstates of S_z are not eigenstates of S_x, S_y too

$\langle\psi|\hat{H}(A\neq 0, B\to 0)|\psi\rangle = ?$

$$S_\pm = S_x \pm i S_y$$

$$S_+^{e^-}S_-^{e^+} = \left(S_x^{e^-} + iS_y^{e^-}\right)\left(S_x^{e^+} - iS_y^{e^+}\right) =$$

$$= S_x^{e^-}S_x^{e^+} - iS_x^{e^-}S_y^{e^+} + iS_y^{e^-}S_x^{e^+} + S_y^{e^-}S_y^{e^+}$$

$$\Rightarrow S_+^{e^-}S_-^{e^+} + S_-^{e^-}S_+^{e^+} = 2\left(S_x^{e^-}S_x^{e^+} + S_y^{e^-}S_y^{e^+}\right)$$

$$A\ \hat{\vec{S}}^{(e^-)} \cdot \hat{\vec{S}}^{(e^+)} =$$

$$= A\left[\frac{1}{2}\left(S_+^{e^-} S_-^{e^+} + S_-^{e^-} S_+^{e^+}\right) + S_z^{e^-} S_z^{e^+}\right]$$

$$\downarrow$$

$$\langle \uparrow_z^{e^-} \downarrow_z^{e^+} | A[\cdots] | \uparrow_z^{e^-} \downarrow_z^{e^+} \rangle =$$

$$= \langle \uparrow_z^{e^-} \downarrow_z^{e^+} | \left\{ 0 + \frac{A}{2}\underbrace{\alpha}_{\text{some real number}} | \downarrow_z^{e^-} \uparrow_z^{e^+} \rangle - A\frac{\hbar^2}{4} | \uparrow_z^{e^-} \downarrow_z^{e^+} \rangle \right\} =$$

$= -A\dfrac{\hbar^2}{4}$, since the spinor eigenstates are orthonormal.